

★ FRACTION + DECIMAL OPERATIONS ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 1 of 12

► Meaning in one line

Add/sub (common.

► Key points

Point	Example
$a/b+c/d = (ad+bc)/bd$; mul/div straight/flip.	$3/4 \div 2/3 = 1/2$; $5/6 \div 6/5 = 1$.
align + count places.	$2.5+0.75 = 3.25$; $1.2 \times 0.4 = 0.48$.

★ IMPORTANT FOR EXAM

Of = multiply; reciprocal + mixed —

⚠ TRAP - read twice

Add-tops + 'bigger-denominator = bigger' + decimal-misalign + no-reduce

✓ **Try yourself:** $3/5+1/10$? 0.6×0.05 ? **Answer:** $7/10$; 0.030 (3 places).

★ CONVERSIONS + COMPARE FRACTIONS ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 2 of 12

► Meaning in one line

Percent/100 , decimal-shift , fraction-reduce , cross-multiply.

► Key points

Point	Example
%/100 + shift + reduce.	37.5% = $\frac{3}{8}$; $0.125 = \frac{1}{8}$; $\frac{2}{3} = 66.67\%$.
ad vs bc; benchmarks $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$.	$\frac{5}{8}$ vs $\frac{7}{11}$: 55 vs 56 → $\frac{7}{11}$ bigger.

★ IMPORTANT FOR EXAM

Terminating ($\frac{2}{5}$!) vs recurring ($\frac{1}{3}$, $\frac{1}{6}$, $\frac{1}{7}$!) + BODMAS

⚠ TRAP - read twice

$37.5\% = 3.75$ + 'bigger-denominator' + no-cross + no-reduce

✓ **Try yourself:** 62.5% as fraction? Bigger: $\frac{4}{7}$ or $\frac{5}{9}$? **Answer:** $\frac{5}{8}$; 36 vs 35 → $\frac{4}{7}$.

★ TERMINATING + BODMAS ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 3 of 12

► Meaning in one line

Denominator 2/5-only = terminating.

► Key points

Point	Example
denominator only 2s/5s.	7/20 terminating; 5/12 recurring (3!).
B-O-DM-AS left-to-right.	$12+8\div2-5 = 12+16-5 = 23$.

★ IMPORTANT FOR EXAM

Brackets + fractional-simplification + approximation

⚠ TRAP - read twice

Left-to-right-all + 'add first' + DM/AS-order

✓ **Try yourself:** Terminating? $9/40$. Value? $20-4\times3+2$. **Answer:** Yes ($40 = 8\times5$); $20-12+2 = 10$.

★ BRACKETS + FRACTIONAL SIMPLIFICATION ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 4 of 12

► Meaning in one line

Inner-first , vinculum , of , cancel , reduce.

► Key points

Point	Example
inner + vinculum + of.	$2+[3\times(4-1)] = 2+9 = 11.$
cancel + reduce.	$(2/3)/(4/9) = 2/3\times9/4 = 3/2.$

★ IMPORTANT FOR EXAM

Approximation + exponents + roots

⚠ TRAP - read twice

Outer-first + of-last + no-cancel + 'simplify = decimal'

✓ **Try yourself:** $5-[2+3\times2]$? Simplify $(5/8)/(15/16)$? **Answer:** $5-8 = -3$; $5/8\times16/15 = 2/3.$

★ APPROXIMATION + EXPONENTS ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 5 of 12

► Meaning in one line

Round , estimate , compatible , 1%/10%.

► Key points

Point	Example
round + compatible + 10%.	$49 \times 21 \approx 50 \times 20 = 1000$ (actual 1029).
add/subtract/multiply powers; $a^0 = 1$.	$2^3 \times 2^7 = 2^{10} = 1024$; $5^0 = 1$.

★ IMPORTANT FOR EXAM

Squares/cubes/roots + surds

⚠ TRAP - read twice

Exact-only + 'approx wrong' + power-add + a0-zero = slow + wrong + fail! Round + rules.

✓ **Try yourself:** $29 \times 12 \approx ?$ Value? $3^2 \times 3^1 = 3^3 = 27$? **Answer:** ≈ 300 (348); $3^1 = 3$.

★ SQUARES/CUBES + SURDS ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 6 of 12

► Meaning in one line

1^2 -30², 1^3 -12³, $\sqrt[3]{\quad}$, perfect, estimate.

► Key points

Point	Example
memorize 1-30 sq + 1-12 cubes.	$17^2 = 289$; $\sqrt{144} = 12$; $\sqrt[3]{27} = 3$.
simplify + rationalize.	$\sqrt{50} = 5\sqrt{2}$; $1/\sqrt{2} = \sqrt{2}/2$.

★ IMPORTANT FOR EXAM

Ratio + proportion

⚠ TRAP - read twice

$\sqrt{(a+b)}$ + 'root distributes' + no-memorize + unrationalized

✓ **Try yourself:** $\sqrt{169} + \sqrt[3]{64}$? Simplify $\sqrt{72}$? **Answer:** $13 + 4 = 17$; $6\sqrt{2}$.

★ RATIO MEANING + EQUIVALENT ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 7 of 12

► Meaning in one line

a:b , parts , total , simplify , divide , scale.

► Key points

Point	Example
a:b = parts; total = sum.	24 in 2:3:4 → 16/3? parts 9: $24 \times \frac{2}{9} \approx 5.33$ —use 27: 6,9,12.
$a/b = c/d \rightarrow ad = bc$.	$x/4 = 9/12 \rightarrow x = 3$.

★ IMPORTANT FOR EXAM

Direct/inverse + division + after-change

⚠ TRAP - read twice

Add-ratio + 'ratio = fraction-value' + no-total/parts

✓ **Try yourself:** 45 in 2:3:5? $x/5 = 12/20$? **Answer:** 9, 13.5→20? 45: 9,13.5,22.5; $x = 3$.

★ DIRECT + INVERSE PROPORTION ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 8 of 12

► Meaning in one line

Direct (up-up.

► Key points

Point	Example
$y = kx.$	5 kg ■ 200 → 8 kg = $8 \times 200 / 5 = 320.$
$xy = k.$	6 men 12 days → 4 men = $6 \times 12 / 4 = 18$ days.

★ IMPORTANT FOR EXAM

Division + after-add + compound

⚠ TRAP - read twice

Direct-always + 'more = direct' + inverse-flip + units-mix

✓ **Try yourself:** Direct: 3 pens 45 → 7? Inverse: 8 pipes 6 h → 12 pipes? **Answer:** 105; $8 \times 6 / 12 = 4$ h.

★ DIVISION IN RATIO + AFTER CHANGE ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 9 of 12

► Meaning in one line

Total $\times$ parts , new-total , new-ratio , difference , constant.

► Key points

Point	Example
share = total $\times$ part/sum.	550 in 3:5:3 $\rightarrow$ 150, 250, 150.
before/after + difference constant.	ratio 4:5, +20 each $\rightarrow$ 6:7? solve: $(4x+20)/(5x+20) = 6/7 \rightarrow x = 20 \rightarrow 80,100.$

★ IMPORTANT FOR EXAM

Compound + percent-bank

⚠ TRAP - read twice

Old-total + 'ratio stays' + difference-ignored

✓ **Try yourself:** 600 in 1:2:3? Ratio 2:3, +10 each $\rightarrow$ 3:4, find? **Answer:** 100,200,300; $(2x+10)/(3x+10)=3/4 \rightarrow x=10 \rightarrow 20,30.$

★ COMPOUND RATIO + PERCENT BANK ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 10 of 12

► Meaning in one line

Multiply , simplify , part-whole.

► Key points

Point	Example
$(a:b)(c:d) = ac:bd$.	$(2:3)(4:5) = 8:15$.
$1/2$ 50, $1/4$ 25, $1/8$ 12.5, $1/3$ 33.33, $2/3$ 66.67, $1/6$ 16.67, $1/7$ 14.28, $1/9$ 11.11, $1/11$ 9.09.	-

★ IMPORTANT FOR EXAM

Basic + increase/decrease

⚠ TRAP - read twice

Add-ratios + 'compound = average' + bank-unmemorized = slow + wrong + fail! Multiply + memorize.

✓ **Try yourself:** $(3:4)(5:7)$? 37.5% + 12.5% as fractions? **Answer:** 15:28; $3/8 + 1/8 = 1/2$.

★ BASIC + INCREASE/DECREASE PERCENT ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 11 of 12

► Meaning in one line

of = multiply , $\text{change/original} \times 100$, new/old.

► Key points

Point	Example
$p\% \text{ of } N = p \times N / 100.$	15% of 240 = 36; 33.33% of 90 = 30.
$\text{change}\% = \text{diff/original} \times 100.$	80→100 = +25%; 120→96 = -20%.

★ IMPORTANT FOR EXAM

Successive + reverse + points

⚠ TRAP - read twice

Percent-of-wrong-base + 'increase = new' + base-mix

✓ **Try yourself:** 20% of 350? 50→65 change%? **Answer:** 70; +30%.

★ SUCCESSIVE + REVERSE PERCENT ★

Quant | Ch02 Fractions + Ratio + Percent | Topic 12 of 12

► Meaning in one line

a, b, $ab/100$, net, chain.

► Key points

Point	Example
$\text{net} = a+b+ab/100$ (signs!).	+20 -10: $20-10-2 = +8\%$.
$O = F \times 100 / (100 \pm r)$.	after +25% = 500 $\rightarrow O = 500 \times 100 / 125 = 400$.

★ IMPORTANT FOR EXAM

Points + population + average

⚠ TRAP - read twice

Add-percent + 'net = sum' + reverse-forward + ab-sign

✓ **Try yourself:** Net: +10 +10? After -20% = 320, original? **Answer:** +21%; $320 \times 100 / 80 = 400$.